

第三章 第一次的作业答案

1、求下列序列 N 点的 DFT

(4) $x(n) = R_m(n) \quad 0 \leq n < N$

$$\begin{aligned} \text{解: } X(k) &= \sum_{n=0}^{N-1} R_m(n) W_N^{kn} = \sum_{n=0}^{m-1} W_N^{kn} = \frac{1 - W_N^{km}}{1 - W_N^k} = \frac{1 - e^{-j\frac{2\pi}{N}km}}{1 - e^{-j\frac{2\pi}{N}k}} = \frac{e^{-j\frac{\pi}{N}km} (e^{j\frac{\pi}{N}km} - e^{-j\frac{\pi}{N}km})}{e^{-j\frac{\pi}{N}k} (e^{j\frac{\pi}{N}k} - e^{-j\frac{\pi}{N}k})} \\ &= e^{-j\frac{\pi}{N}(m-1)k} \frac{\sin\left(\frac{\pi}{N}mk\right)}{\sin\left(\frac{\pi}{N}k\right)} \quad k = 0, 1, \dots, N-1 \end{aligned}$$

(6) $x(n) = \cos\left(\frac{2\pi}{N}mn\right)$

$$\text{解: } X(k) = \sum_{n=0}^{N-1} \cos\left(\frac{2\pi}{N}mn\right) W_N^{kn} = \sum_{n=0}^{N-1} \frac{1}{2} \left[e^{j\frac{2\pi}{N}mn} + e^{-j\frac{2\pi}{N}mn} \right] e^{-j\frac{2\pi}{N}kn} = \frac{1}{2} \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(m-k)n} + \frac{1}{2} \sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}(m+k)n}$$

$$= \frac{1}{2} \frac{1 - e^{j\frac{2\pi}{N}(m-k)N}}{1 - e^{j\frac{2\pi}{N}(m-k)}} + \frac{1}{2} \frac{1 - e^{-j\frac{2\pi}{N}(m+k)N}}{1 - e^{-j\frac{2\pi}{N}(m+k)}} = \frac{1}{2} \frac{1 - e^{j2\pi(m-k)}}{1 - e^{j\frac{2\pi}{N}(m-k)}} + \frac{1}{2} \frac{1 - e^{-j2\pi(m+k)}}{1 - e^{-j\frac{2\pi}{N}(m+k)}}$$

$$= \begin{cases} \frac{N}{2} & \text{当 } k = m \text{ 和 } k = N - m \text{ 时} \\ 0 & \text{其他 } k \text{ 值} \end{cases} = \frac{N}{2} \delta(k - m) + \frac{N}{2} \delta(k - (N - m))$$

2、已知 $X(k)$, 求 $x(n) = \text{IDFT}[X(k)]$

$$X(k) = \begin{cases} \frac{N}{2} e^{j\theta} & k = m \\ \frac{N}{2} e^{-j\theta} & k = N - m \\ 0 & \text{其他 } k \text{ 值} \end{cases}$$

$$\begin{aligned} \text{解: } x(n) &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} \left[\frac{N}{2} e^{j\theta} \delta(k - m) + \frac{N}{2} e^{-j\theta} \delta(k - N + m) \right] W_N^{-kn} \\ &= \frac{1}{N} \left[\frac{N}{2} e^{j\theta} W_N^{-mn} + \frac{N}{2} e^{-j\theta} W_N^{-(N-m)n} \right] = \frac{1}{2} \left[e^{j\theta} e^{j\frac{2\pi}{N}mn} + e^{-j\theta} e^{-j\frac{2\pi}{N}mn} \right] \\ &= \frac{1}{2} \left[e^{j\left(\theta + \frac{2\pi}{N}mn\right)} + e^{-j\left(\theta + \frac{2\pi}{N}mn\right)} \right] = \cos\left(\theta + \frac{2\pi}{N}mn\right) \quad n = 0, 1, \dots, N-1 \end{aligned}$$

3、已知长度 $N=10$ 的两个序列，用矩阵相乘法计算两序列 $L=10$ 点长的循环卷积。

$$x_1(n) = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & 5 \leq n \leq 9 \end{cases}, \quad x_2(n) = \begin{cases} 1 & 0 \leq n \leq 4 \\ -1 & 5 \leq n \leq 9 \end{cases}$$

$$\text{解: } y(n) = x_1(n) \circledast x_2(n) = \left[\sum_{m=0}^9 x_2(m) x_1((n-m))_{10} \right] R_{10}(n)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \\ y(5) \\ y(6) \\ y(7) \\ y(8) \\ y(9) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \\ 5 \\ 3 \\ 1 \\ -1 \\ -3 \\ -5 \end{bmatrix}$$

$$\begin{aligned} \text{DFT 对偶性: } x(n) &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \\ n \leftrightarrow k \quad Nx(k) &= \sum_{n=0}^{N-1} X(n) W_N^{-kn} \\ Nx(N-k) &= \sum_{n=0}^{N-1} X(n) W_N^{-(N-k)n} \\ &= \sum_{n=0}^{N-1} X(n) W_N^{kn} \\ \therefore X(n) &\leftrightarrow Nx(N-k) \end{aligned}$$

4、证明 DFT 的对称性，即假设 $X(k) = \text{DFT}[x(n)]$ ，证明 $\text{DFT}[X(n)] = Nx(N-k)$ 。

$$\text{证明: } X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{m=0}^{N-1} x(m) W_N^{km}$$

$$\text{DFT}[X(n)] = \sum_{n=0}^{N-1} X(n) W_N^{kn} = \sum_{n=0}^{N-1} \left[\sum_{m=0}^{N-1} x(m) W_N^{nm} \right] W_N^{kn} = \sum_{m=0}^{N-1} x(m) \sum_{n=0}^{N-1} W_N^{(k+m)n}$$

$$\text{因为: } \sum_{n=0}^{N-1} W_N^{(k+m)n} = \frac{1 - W_N^{(k+m)N}}{1 - W_N^{k+m}} = \frac{1 - e^{-j\frac{2\pi}{N}(k+m)N}}{1 - e^{-j\frac{2\pi}{N}(k+m)}} = \frac{1 - e^{-j2\pi(k+m)}}{1 - e^{-j\frac{2\pi}{N}(k+m)}} = \begin{cases} N & k+m = N \\ 0 & k+m \neq N \end{cases}$$

$$\text{所以: } \text{DFT}[X(n)] = Nx(N-k)$$

6、设 $x(n)$ 的长度为 N ，且 $X(k) = \text{DFT}[x(n)]$ $0 \leq k \leq N-1$ ，令 $h(n) = x((n))_N R_{mN}(n)$ ， $H(k) = \text{DFT}[h(n)]|_{mN} \ 0 \leq k \leq mN-1$ ，求 $H(k)$ 与 $X(k)$ 的关系。

$$\text{解: } H(k) = \sum_{n=0}^{mN-1} h(n) W_{mN}^{kn} = \sum_{n=0}^{mN-1} x((n))_N R_{mN}(n) W_{mN}^{kn} = \sum_{n=0}^{mN-1} x((n))_N W_{mN}^{kn} \quad \text{设 } n = n' + lN$$

$$= \sum_{l=0}^{m-1} \sum_{n'=0}^{N-1} x((n'+lN))_N W_{mN}^{k(n'+lN)} = \sum_{l=0}^{m-1} \sum_{n'=0}^{N-1} x(n') W_{mN}^{kn'} W_{mN}^{klN} = \sum_{l=0}^{m-1} \sum_{n'=0}^{N-1} \left[x(n') W_N^{\frac{k}{m}n'} \right] W_m^{kl} = X\left(\frac{k}{m}\right) \sum_{l=0}^{m-1} W_m^{kl}$$

$$\text{由于: } \sum_{l=0}^{m-1} W_m^{kl} = \sum_{l=0}^{m-1} e^{-j\frac{2\pi}{m}kl} = \frac{1 - e^{-j\frac{2\pi}{m}km}}{1 - e^{-j\frac{2\pi}{m}k}} = \frac{1 - e^{-j2\pi k}}{1 - e^{-j\frac{2\pi}{m}k}} = \begin{cases} m & \frac{k}{m} \text{ 为常数} \\ 0 & \frac{k}{m} \text{ 不为常数} \end{cases}$$

$$H(k) = \begin{cases} mX\left(\frac{k}{m}\right) & \frac{k}{m} \text{ 为常数} \\ 0 & \frac{k}{m} \text{ 不为常数} \end{cases}$$

7. 证明: 若 $x(n)$ 实奇对称 即 $x(n) = -x(N-n)$, 则 $X(k)$ 为纯虚函数并奇对称。

证明:
$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$X(k) = \sum_{n=0}^{N-1} -x(N-n) W_N^{kn} \stackrel{N-n=m}{=} \sum_{m=N}^1 -x(m) W_N^{k(N-m)} = \sum_{n=0}^{N-1} -x(n) W_N^{-kn}$$

$$X(k) = - \sum_{n=0}^{N-1} x(n) W_N^{kn} = - \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad (x(n) \text{ 为实})$$

$$\therefore X(k) = -X^*(k) \quad \text{Re}[X(k)] + j\text{Im}[X(k)] = -\text{Re}[X(k)] + j\text{Im}[X(k)]$$

$$\therefore \text{Re}[X(k)] = 0 \quad \therefore X(k) \text{ 为纯虚函数}$$

$$\therefore \bar{X}(N-k) = \sum_{n=0}^{N-1} x(n) W_N^{(N-k)n} = \sum_{n=0}^{N-1} x(n) W_N^{-kn} = -X(k)$$

$$\therefore X(k) \text{ 奇对称。}$$

9. 已知 $x(n)$ 的周期为 N . $X(k) = \text{DFT}[x(n)]$, $y(n) = \begin{cases} x(n) & 0 \leq n \leq N-1 \\ 0 & N \leq n \leq mN-1 \end{cases}$

$Y(k) = \text{DFT}[y(n)]_{mN}$ 求 $Y(k)$ 与 $X(k)$ 的关系。

证:
$$Y(k) = \sum_{n=0}^{mN-1} y(n) W_{mN}^{kn} \quad k=0, 1, \dots, mN-1$$

$$= \sum_{n=0}^{N-1} x(n) W_{mN}^{kn} = \sum_{n=0}^{N-1} x(n) W_N^{\frac{k}{m}n}$$

$$= \begin{cases} X(\frac{k}{m}) & \frac{k}{m} \text{ 为整数时} \\ \sum_{n=0}^{N-1} x(n) W_N^{\frac{k}{m}n}, & \frac{k}{m} \text{ 不为整数时} \end{cases}$$

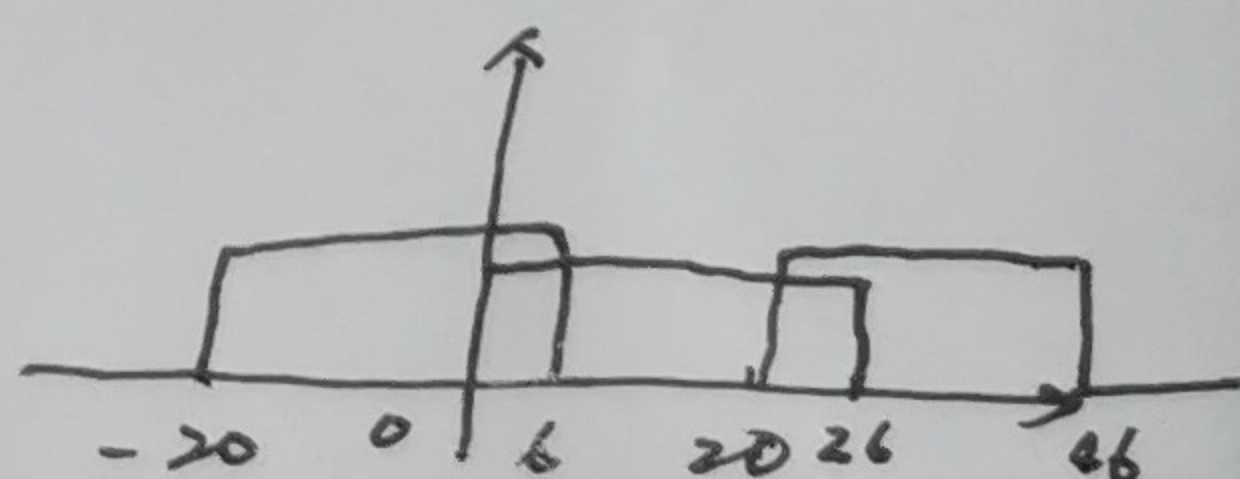
$$Y(k) = X(\frac{k}{m}) \quad \frac{k}{m} \text{ 为整数}$$

14. 两有限长序列 $x(n)$ 和 $y(n)$ 的零值区间为: $x(n)=0, n < 0, 8 \leq n$; $y(n)=0, n < 0, 20 \leq n$, 对每个序列作 20 点 DFT, 即 $X(k) = \text{DFT}[x(n)]_{20}$, $Y(k) = \text{DFT}[y(n)]_{20}$, 如果 $F(k) = X(k) \cdot Y(k)$, $f(n) = \text{IDFT}[F(k)]_{20}$, 试问在哪些点上 $f(n)$ 与 $x(n) * y(n)$ 值相等. 为什么?

证: $x(n)$ 的长 $N=8$. $y(n)$ 的长 $M=20$. $f_2(n) = x(n) * y(n)$ 的长为 $20+8-1=27$

$$f(n) = x(n) \circledast y(n) = f_R(n)_{20} R_{20}(n)$$

在 $n=7 \sim 19$ 点上. $f(n)$ 与 $x(n) * y(n)$ 的值相等。



15. 已知实序列 $x(n)$ 的 8 点 DFT 的 5 个值为 $0.25, 0.125 - j0.3018, 0, 0.125 + j0.0518, 0$,
 (1) 求 $X(k)$ 的其余 3 个值; (2) $x_1(n) = \sum_{m=-\infty}^{\infty} x(n+5+8m) R_8(n)$, 求 $X_1(k) = \text{DFT}[x_1(n)]$
 (3) $x_2(n) = x(n) e^{j\pi n/4}$, 求 $X_2(k) = \text{DFT}[x_2(n)]$ 。

解: (1) $X(5) = X(8-5) = X(3)^* = 0.125 + j0.0518$
 $X(6) = X(8-6) = X(2)^* = 0$
 $X(7) = X(8-7) = X(1)^* = 0.125 + j0.3018$

(2) $x_1(n) = x((n+5))_8 R_8(n) \quad \therefore X_1(k) = W_8^{-5k} X(k)$

(3) $x_2(n) = W_8^n x(n) \leftrightarrow X_2(k) = X((k+1))_8 R_8(k) = \{ \underline{X(7)}, X(0), X(1), X(2), X(3), X(4), X(5), X(6) \}$

18. 用微处理器对实序列作谱分析, 要求谱分辨率 $F \leq 50$, 信号最高频率为 1KHz , 确定以下各参数; (1) T_{pmin} ; (2) T_{max} ; (3) N_{min} (4) 求谱分辨率提高 1 倍在 N 值。

解: (1) $T_{pmin} = \frac{1}{F_{max}} = \frac{1}{50} = 0.02\text{ s}$

(2) $T_{max} = \frac{1}{f_{smin}} = \frac{1}{f_m} = \frac{1}{2 \times 10^3} = 0.5\text{ ms}$

(3) $N_{min} = \frac{T_{pmin}}{T_{max}} = \frac{0.02}{0.5 \times 10^{-3}} = 40$

(4) $F \leq 25\text{ Hz} \quad T_{pmin} = \frac{1}{25} \quad N_{min} = \frac{1}{0.5 \times 10^{-3}} = 80$